

Partitioning changes in ecosystem productivity by effects of species interactions in biodiversity experiments

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eLife Assessment

The authors propose that positive biodiversity-ecosystem functioning relationships found in experiments have been exaggerated because commonly used statistical analyses are flawed. To remedy this, a new type of analysis based on a concept of "partial density monoculture yield" is proposed. However, the presented concept and analysis methods are not reproducibly described, do not appear to be complete, and are **inadequate** for hypothesis testing. The reviewers found that the authors misinterpret current research in the field and made limited efforts to understand or address the reviewer comments on a previous version of the study.

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Abstract

Species interactions affect ecosystem productivity. Positive interactions (resource partitioning and facilitation) increase productivity while negative interactions (species interference) decrease productivity relative to the null expectations defined by monoculture yields. Effects of competitive interactions (resource competition) can be either positive or negative. Distinguishing effects of species interactions is therefore difficult, if not impossible, with current biodiversity experiments involving mixtures and full density monocultures.

To partition changes in ecosystem productivity by effects of species interactions, we modify null expectations with competitive growth responses, i.e., proportional changes in individual size (biomass or volume) expected in mixture based on species differences in growth and

competitive ability. We use partial density (species density in mixture) monocultures and the competitive exclusion principle to determine maximum competitive growth responses and full density monoculture yields to measure species ability to achieve maximum competitive growth responses in mixture. Deviations of observed yields from competitive expectations represent the effects of positive/negative species interactions, while the differences between competitive and null expectations reflect the effects of competitive interactions.

We demonstrate the effectiveness of our competitive partitioning model in distinguishing effects of species interactions using both simulated and experimental species mixtures. Our competitive partitioning model enables meaningful assessments of species interactions at both species and community levels and helps disentangle underlying mechanisms of species interactions responsible for changes in ecosystem productivity and identify species mixtures that maximize positive effects.

Species interactions and ecosystem productivity

Detecting changes in ecosystem productivity with biodiversity and understanding how these changes are affected by mechanisms of species interactions has been a central focus in ecology for decades (Loreau, 2000 [DOI](#); Loreau and Hector, 2019 [DOI](#); Mahaut et al., 2020 [DOI](#)). However, effective methods are lacking (Barry et al., 2019 [DOI](#); Bourrat et al., 2023 [DOI](#); Drake, 2003 [DOI](#); Forrester and Pretzsch, 2015 [DOI](#); Jaillard et al., 2018 [DOI](#); Loreau and Hector, 2019 [DOI](#)). The method of additive partitioning was developed from the covariance equation to quantify changes in ecosystem productivity by comparing species performance in mixture with those expected from their monocultures using a replacement design (Barry et al., 2019 [DOI](#); Loreau and Hector, 2001 [DOI](#)). The resultant changes in ecosystem productivity (commonly referred to as net biodiversity effects) are mathematically decomposed into additive components, namely the complementarity effect (CE) measuring community average response and the selection effect (SE) measuring interspecies differences (Forrester and Pretzsch, 2015 [DOI](#); Loreau and Hector, 2001 [DOI](#), 2019). CE and SE are widely used to examine mechanisms of species interactions responsible for biodiversity effects, i.e., positive CE for positive interactions and positive SE for competitive interactions (Barry et al., 2019 [DOI](#); Feng et al., 2022 [DOI](#); Hagan et al., 2021 [DOI](#); Loreau and Hector, 2001 [DOI](#); Montès et al., 2008 [DOI](#); Petchey, 2003 [DOI](#); Polley et al., 2003 [DOI](#)), despite that neither of the additive components can be linked to specific effects of species interactions (Bourrat et al., 2023 [DOI](#); Forrester and Pretzsch, 2015 [DOI](#); Hagan et al., 2021 [DOI](#); Hooper et al., 2005 [DOI](#); Loreau et al., 2012 [DOI](#); Loreau and Hector, 2019 [DOI](#); Montès et al., 2008 [DOI](#); Petchey, 2003 [DOI](#)).

Among the interspecific interactions in plants, positive interactions (resource partitioning and facilitation or collectively referred to as complementarity) increase productivity and negative interactions (species interference or interference competition) decrease productivity relative to the null expectations, i.e., species yields expected in mixture based on composition (relative yield) and full density monoculture yields (Barry et al., 2019 [DOI](#); Hooper et al., 2005 [DOI](#); Loreau, 2000 [DOI](#); Loreau and Hector, 2001 [DOI](#)). Uncertainty comes with effects of competitive interactions (i.e., resource competition) that are positive for more competitive species and negative for less competitive species (Aschehoug et al., 2016 [DOI](#); Barry et al., 2019 [DOI](#)). For example, when a highly competitive and less competitive species are mixed, more resources flow to the more competitive species. The yield gain of more competitive species from increases in resource availability (resources per capita) often exceeds the yield loss of less competitive species from decreases in resource availability (Huston, 1997 [DOI](#); Montazeaud et al., 2018 [DOI](#); Pillai and Gouhier, 2019 [DOI](#); Tilman et al., 1997 [DOI](#); Wardle, 1999 [DOI](#)), resulting in positive deviation from the total expected yield of both species (i.e., positive biodiversity effect). Competitive interactions can also result in negative biodiversity effects if the less competitive species suffers more than the more competitive species benefits.

The positive contribution of competitive dominance to ecosystem productivity based on the null expectation can be illustrated with species mixture data. In the situations of both simulated and experimental mixtures with two species (Huang et al., 2009 [↗](#); Mahaut et al. 2020 [↗](#)), the partial density monoculture yield of the more competitive species exceeded the total expected yield of both species in 32 of the 35 mixtures (**Table 1** [↗](#)), meaning that a higher productivity expected for a given species mixture under the null hypothesis can be generally achieved by removing less competitive species from the mixture. In other words, the positive biodiversity effects of mixtures detected through the null expectation can be obtained from the monocultures of more competitive species at partial (lower) densities. These positive biodiversity effects, however, result from competition-induced changes in resource availability, i.e., the yield gain of more competitive species from all resources in the mixture exceeds the total yield loss expected for the less competitive species, and are not related to effects of positive species interactions. Competitive interactions are the predominant type of interspecific relationships in plants and occur in all mixtures where constituent species are competitively different (Goldberg and Barton, 1992 [↗](#); Pillai and Gouhier, 2019 [↗](#)). Reporting positive biodiversity effects, without indication of possible contributions from competitive interactions, can lead to unrealistic expectations for effects of positive interactions.

Given the limits of the additive partitioning in deciphering mechanisms of changes in ecosystem productivity (Forrester and Pretzsch, 2015 [↗](#); Loreau and Hector, 2001 [↗](#); Petchey, 2003 [↗](#)), some alternative methods have been explored to identify underlying mechanisms. This includes the use of metrics such as relative yield totals (Hector, 1998 [↗](#); Roscher and Schumacher, 2016 [↗](#)) and transgressive overyielding or superyielding (Drake, 2003 [↗](#); Forrester and Pretzsch, 2015 [↗](#); Špačková and Lepš, 2001 [↗](#)), the multiplicative partitioning or statistical analyses to separate effects of species interactions and species composition (Chen et al., 2020 [↗](#); Jaillard et al., 2018 [↗](#); Kirwan et al., 2009 [↗](#)), the identification of relative species influences via measurement of competitive interactions (Brun et al., 2022 [↗](#); Mahaut et al., 2020 [↗](#); Montès et al., 2008 [↗](#)), or influences of species diversity, composition, and density on net biodiversity effects (Jaillard et al., 2018 [↗](#); Mahaut et al., 2020 [↗](#); Polley et al., 2003 [↗](#); Roscher and Schumacher, 2016 [↗](#); Tatsumi and Loreau, 2023 [↗](#)). However, although these methods may indicate the presence of strong positive or competitive interactions or connections of the additive components with niche and fitness differences (Carroll et al., 2011 [↗](#); Godoy et al., 2020 [↗](#); Turnbull et al., 2013 [↗](#)), they are not capable of quantifying changes in ecosystem productivity by species interactions at species or community level (Loreau et al., 2012 [↗](#)).

Here, we present the competitive partitioning model, a new framework for assessing effects of species interactions on ecosystem productivity based on null expectations and competitive growth responses, the proportional changes of individual size (biomass or volume) from full density monocultures to mixtures expected from species differences in growth and competitive ability (**Figure 1** [↗](#)). Current biodiversity experiments are generally established based on a replacement design with equal density across mixtures and monocultures (Sackville Hamilton, 1994 [↗](#)) and do not provide data for estimating competitive growth responses. Thus, we applied this new methodological approach to simulated data generated with the GYPsy model (Huang et al., 2009 [↗](#)) for trembling aspen (*Populus tremuloides* Michx.) and white spruce (*Picea glauca* (Moench) Voss) tree mixtures and greenhouse experimental data from grassland mixtures (Mahaut et al., 2020 [↗](#)) (**Table 1** [↗](#)). We used partial density (species density in mixture) monocultures and the competitive exclusion principle to determine maximum competitive growth responses and full density monoculture yields to measure species ability to compete for resources and achieve maximum competitive growth responses in mixture. Our objectives are to (1) demonstrate the power of this new framework in distinguishing effects of species interactions responsible for changes in ecosystem productivity, (2) illustrate how the mechanisms of changes in ecosystem productivity differ between the assessments with competitive and additive partitioning models, and (3) assess the effects of competitive interactions on ecosystem productivity relative to those of other species interactions.

Table 1.

Simulated and experimental species mixtures. Biodiversity effects with partial density monocultures (BPM) are calculated from the differences between the partial density monoculture yield of more competitive species and total mixture yield expected from species relative yields and full density monoculture yields (see competitive exclusion in Tables S1 and S2 for detailed calculations).

Species mixture (species1:species2)	Age	Total density	Mixture composition		Full density monoculture yields		Observed yields in mixtures		Partial density monoculture yields		
			Species1	Species2	Species1	Species2	Species1	Species2	Species1	Species2	BPM
Simulated mixed trembling aspen and white spruce in western Canada generated with GYPSY model (<i>Huang et al., 2009</i>) (density and stand volume yield, stems and m³ per hectare)											
Populus:Picea	20 years	11,000	0.9	0.1	23.7	0.7	22.3	0.1	23.3	0.5	1.9
Populus:Picea	20 years	11,000	0.7	0.3	23.7	0.7	19.0	0.2	22.5	0.6	5.7
Populus:Picea	20 years	11,000	0.5	0.5	23.7	0.7	15.1	0.3	21.4	0.6	9.2
Populus:Picea	20 years	11,000	0.3	0.7	23.7	0.7	10.6	0.4	19.8	0.7	12.2
Populus:Picea	20 years	11,000	0.1	0.9	23.7	0.7	5.0	0.5	16.9	0.7	13.9
Populus:Picea	40 years	3,600	0.9	0.1	110.4	27.2	104.5	3.4	108.3	15.5	6.3
Populus:Picea	40 years	3,600	0.7	0.3	110.4	27.2	90.3	7.6	103.5	20.7	18.1
Populus:Picea	40 years	3,600	0.5	0.5	110.4	27.2	72.4	12.1	97.0	23.4	28.2
Populus:Picea	40 years	3,600	0.3	0.7	110.4	27.2	49.6	17.5	87.4	25.3	35.2
Populus:Picea	40 years	3,600	0.1	0.9	110.4	27.2	20.6	23.9	70.6	26.7	35.1
Populus:Picea	60 years	1,800	0.9	0.1	189.7	88.3	180.5	9.6	186.1	47.0	6.6
Populus:Picea	60 years	1,800	0.7	0.3	189.7	88.3	158.1	22.4	177.3	64.3	18.0
Populus:Picea	60 years	1,800	0.5	0.5	189.7	88.3	128.1	38.1	164.8	73.7	25.8
Populus:Picea	60 years	1,800	0.3	0.7	189.7	88.3	85.4	58.5	144.8	80.5	26.1
Populus:Picea	60 years	1,800	0.1	0.9	189.7	88.3	28.6	83.0	109.2	85.9	10.8
Populus:Picea	80 years	1,100	0.9	0.1	240.2	143.1	229.6	13.7	235.6	73.3	5.1
Populus:Picea	80 years	1,100	0.7	0.3	240.2	143.1	203.7	33.0	224.1	102.0	13.1
Populus:Picea	80 years	1,100	0.5	0.5	240.2	143.1	167.3	58.2	207.6	117.8	15.9
Populus:Picea	80 years	1,100	0.3	0.7	240.2	143.1	110.3	94.0	179.1	129.4	6.8
Populus:Picea	80 years	1,100	0.1	0.9	240.2	143.1	26.2	139.6	124.5	138.9	-28.4
Experimental grassland mixtures (density and aboveground biomass yield, stems and grams per pot) (<i>Mahaut et al., 2020</i>)											
Bromus: Dactylis	13 weeks	6	0.5	0.5	5.5	9.8	2.6	8.6	5.3	11.5	3.8
Bromus: Lotus	13 weeks	6	0.5	0.5	5.5	6.3	3.8	3.2	5.3	4.7	-1.2
Bromus:Plantago	13 weeks	6	0.5	0.5	5.5	8.6	1.9	10.9	5.3	16.1	9.0
Bromus:Sanguisorba	13 weeks	6	0.5	0.5	5.5	4.4	2.9	2.2	5.3	3.6	0.3
Bromus:Trifolium	13 weeks	6	0.5	0.5	5.5	13.3	3.8	9.3	5.3	14.2	4.8
Dactylis:Lotus	13 weeks	6	0.5	0.5	9.8	6.3	11.8	2.0	11.5	4.7	3.4
Dactylis:Plantago	13 weeks	6	0.5	0.5	9.8	8.6	4.4	9.5	11.5	16.1	2.3
Dactylis:Sanguisorba	13 weeks	6	0.5	0.5	9.8	4.4	6.0	1.9	11.5	3.6	4.4
Dactylis:Trifolium	13 weeks	6	0.5	0.5	9.8	13.3	7.5	7.7	11.5	14.2	2.6
Lotus:Plantago	13 weeks	6	0.5	0.5	6.3	8.6	1.0	16.2	4.7	16.1	8.6
Lotus:Sanguisorba	13 weeks	6	0.5	0.5	6.3	4.4	1.9	2.6	4.7	3.6	-0.6
Lotus:Trifolium	13 weeks	6	0.5	0.5	6.3	13.3	2.4	9.2	4.7	14.2	4.4
Plantago:Sanguisorba	13 weeks	6	0.5	0.5	8.6	4.4	11.6	1.1	16.1	3.6	9.6
Plantago:Trifolium	13 weeks	6	0.5	0.5	8.6	13.3	10.2	6.9	16.1	14.2	3.3
Sanguisorba:Trifolium	13 weeks	6	0.5	0.5	4.4	13.3	1.9	13.4	3.6	14.2	5.3

Species codes: Populus - *Populus tremuloides*, Picea - *Picea glauca*, Bromus - *Bromus erectus*, Dactylis - *Dactylis glomerata*, Lotus - *Lotus corniculatus*, Plantago - *Plantago lanceolata*, Sanguisorba - *Sanguisorba minor*, and Trifolium - *Trifolium repens*.

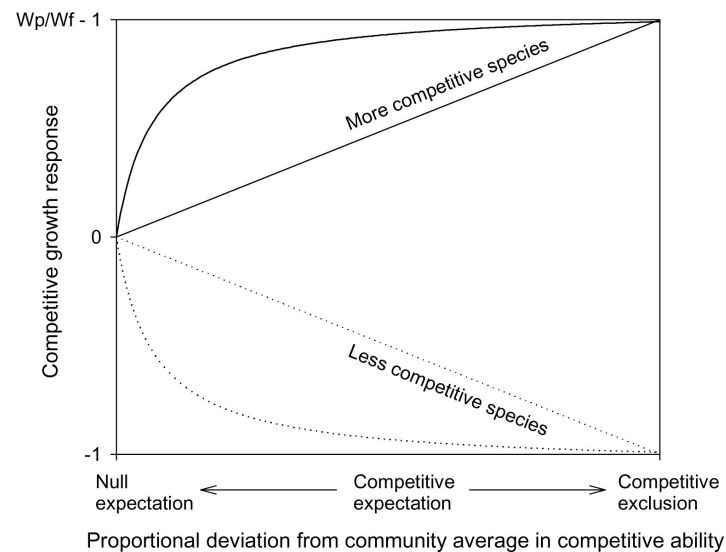


Figure 1.

Under the competitive hypothesis, more competitive species gain size (biomass or volume) and less competitive species lose size in mixture relative to their full density monocultures. The magnitude of competitive growth responses (i.e., proportional changes in individual size) increases with relative competitive ability (proportional deviations of species competitive ability from community average) from a minimum of 0 (at community average or null expectation) to a maximum of $W_p/W_f - 1$ for more competitive species and -1 for less competitive species at competitive exclusion. The changes may not be linear, greater near community average and smaller with deviations of competitive ability (Gaudet and Keddy, 1988) or resource availability (e.g., light, see Brüllhardt et al., 2020) away from community averages. W_f represents for individual size in full density monocultures and W_p represents for individual size in partial density monocultures.

Competitive expectation

Under the null hypothesis, species in mixture are assumed to be competitively equivalent (i.e., equal interspecific and intraspecific interactions) and species growth (or any other function) stays the same in mixture as in monocultures (Loreau and Hector, 2001 [\[1\]](#)). Deviations of observed yields (biomass or volume) from total null expectation therefore produce net biodiversity effects or changes in ecosystem productivity (Barry et al., 2019 [\[2\]](#); Loreau and Hector, 2001 [\[1\]](#)).

$$\begin{aligned}\sum Y_{Oi} - \sum Y_{NEi} &= \sum RY_{Oi}M_{if} - \sum RY_{Ei}M_{if} \\ &= \sum \Delta RY_i M_{if}\end{aligned}\quad (1)$$

Thus, the net biodiversity effect is the difference between total observed yields $\sum Y_{Oi}$ and total null expectation $\sum Y_{NEi}$, calculated from species composition (or relative yield) in mixture RY_{Ei} and full density monoculture yield M_{if} .

To separate effects of competitive interactions from those of other species interactions, we would need the hypothesis that constituent species share an identical niche but differ in growth and competitive ability (i.e., absence of positive/negative interactions). Competitive growth responses, i.e., proportional changes in individual size (biomass or volume) expected from full density monocultures to mixtures under this competitive hypothesis, can be determined from species differences in growth and competitive ability (**Figure 1** [\[3\]](#)). The magnitudes of competitive growth responses depend on: (1) species resource needs in full density monocultures; and (2) changes in resource availability (resources per capita) from full density monocultures to full density mixtures that are related to species composition and relative ability to compete for resources in mixture.

To estimate competitive growth response, we first determine the maximum competitive growth response MG_i that species can reach in a scenario of competitive exclusion (Murray, 1986 [\[4\]](#)) where a more competitive species completely dominates and grows like a partial density monoculture, while a less competitive species is competitively eliminated.

$$\begin{aligned}MG_i &= \frac{w_{ip}}{w_{if}} - 1, \text{ for more competitive species} \\ &= -1, \text{ for less competitive species}\end{aligned}\quad (2)$$

Here, w_{ip} stands for individual size (biomass or volume) in partial density (species density in mixture) monoculture and w_{if} is for individual size in full density monoculture; both parameters are calculated from species monoculture yields and initial densities in partial/full density monocultures. A more competitive species gains $\frac{1}{RY_{Ei}} - 1$ in resource availability and $\frac{w_{ip}}{w_{if}} - 1$ in individual size, while a less competitive species loses 100% resource availability and size in mixture relative to full density monoculture (**Figure 1** [\[3\]](#)). This simplified competitive dominance applies to mixtures where constituent species differ considerably in competitive ability, especially when competition for light is intense (Grime, 1973 [\[5\]](#); Lieffers et al., 1993 [\[6\]](#)). In most cases, however, competitive exclusion would not occur in biodiversity experiments that are controlled in density, composition, duration, and scale. More competitive species are negatively affected by interspecific competition relative to their partial density monocultures (Holland and DeAngelis, 2009 [\[7\]](#)), while less competitive species are competitively suppressed relative to their null expectations, but not eliminated (Aarssen, 1989 [\[8\]](#); Connolly et al., 2001 [\[9\]](#); Mahaut et al., 2020 [\[10\]](#); Montazeaud et al., 2018 [\[11\]](#)). Other than changes in resource availability, the estimated maximum competitive growth response would include the effects of density-dependent pests, pathogens, or microclimates.

Next, we use relative competitive ability RC_i to represent species ability to compete for resources and achieve their maximum competitive growth response in mixture (Aarssen, 1989). RC_i is defined by the deviation of species competitive ability H_i from community average $\bar{H}_{i \notin}$ (excluding species i) relative to community maximum H_m . Competitive ability is species ability to exploit resources and can be assessed by growth attributes such as full density monoculture yields in biomass or volume (Aarssen, 1989; Aschehoug et al., 2016; Gaudet and Keddy, 1988; Grace, 1990; Montazeaud et al., 2018).

$$RC_i = \frac{|H_i - \bar{H}_{i \notin}|}{H_m} \quad (3)$$

Greater competitive advantages would lead to increasingly positive competitive growth responses and larger competitive disadvantages would lead to increasingly negative competitive growth responses in mixture relative to full density monocultures.

$$Y_{CEi} = Y_{NEi} [1 + MG_i RC_i] \quad (4)$$

Species competitive expectation Y_{CEi} is the yield based on null expectation Y_{NEi} and competitive growth response $MG_i RC_i$ derived from species differences in growth and competitive ability. The term $MG_i RC_i$ would be positive for more competitive species $MG_i > 0$, negative for less competitive species $MG_i < 0$, and zero when species are at community average or competitively equivalent $RC_i = 0$ (Figure 1). At individual species level, the deviation of observed yield Y_{Oi} from the competitive expectation Y_{CEi} can be positive (representing relaxation of interspecific competition from competitive expectation and therefore dominance of positive interactions), neutral (representing similar intra- and interspecific interactions or offset of positive and negative interactions), or negative (representing dominance of negative interactions). The difference between competitive and null expectations is competitive yield change resulting from competitive interactions. Isolating facilitative effect from those of other positive interactions (i.e., resource partitioning) is also possible for individual species by comparing species' observed yield in mixture to their partial density monoculture yield, adapting the principle of additive design for biodiversity–ecosystem productivity experiments (Sackville Hamilton, 1994). At the community level, the difference in total yield between competitive expectations $\sum Y_{CEi}$ and null expectations $\sum Y_{NEi}$ of all species can be attributed to the competitive effect, whereas the difference between observed yields $\sum Y_{Oi}$ and competitive expectations $\sum Y_{CEi}$ can be attributed to the dominance of positive effects when the difference > 0 or to the dominance of negative effects when the difference < 0 . The community positive effect can be further partitioned by mechanisms of positive interactions (resource partitioning and facilitation), and facilitative effect can be classified as mutualism (+/+), commensalism (+/0), or parasitic (+/−) based on species specific assessments.

Case studies

Simulated aspen and spruce mixtures

Mixed forests composed of trembling aspen (*Populus tremuloides* Michx.) and white spruce (*Picea glauca* (Moench) Voss) are widely distributed in North America (Man and Lieffers, 1999). The two species, occurring in pure and mixed stands of varying compositions in natural conditions, generally have different canopy structure, rooting depth, leaf phenology, and requirements for light, moisture, and nutrients so are considered different in niche (Man and Lieffers, 1999).

We used a growth and yield simulation system GYPsy, developed for mixed species forests in western Canada (Huang et al., 2009), to generate data required for determining competitive expectation, i.e., mixtures and monocultures at partial and full densities. We were interested in knowing how effects of species interactions may vary with stand age and composition. We chose

four ages: 20, 40, 60, and 80 years, to represent young, middle-aged, pre-mature, and mature stands, and five relative yields from nearly pure aspen (90% aspen), aspen-dominated (70% aspen), equal mixed (50% aspen), spruce-dominated (30% aspen), to nearly pure spruce (10% aspen) mixtures at average densities on fully stocked medium productivity sites (Peterson and Peterson, 1992) (Table 1). As GYPSY is developed from extensive field observations (Huang et al., 2009), the simulated data represent average yields in natural monocultures and mixtures of various composition and ages that may result from direct or indirect interactions between the two species (Barry et al., 2019; Man and Lieffers, 1999). Species competitive ability was approximated with stand volume in full density monocultures at site index 17 m and 14 m (height), respectively, for dominant and codominant aspen and spruce at 50 years of total age on medium productivity sites (Alberta Forest Service, 1985; Huang et al., 1994).

The partitioning of net biodiversity effects with competitive partitioning model indicated that the relative contribution of competitive interactions to net biodiversity effects decreased with age (Figure 2a), whereas community positive effects occurred generally in pre-mature and mature stands and more so in equal aspen and spruce mixtures (Figure 2b). Evidence for a facilitative effect (i.e., species observed yield in mixture > partial density monoculture yield) was only found on spruce in nearly pure mature spruce mixture (Table 1). Community negative effects occurred in young and middle-aged mixtures where aspen did poorer than expected under the competitive expectation (Table S1). Averaged across all ages and compositions, the net biodiversity effect was $14.3 \text{ m}^3 \text{ ha}^{-1}$ or 14% of the average full density monoculture yield. On a relative basis, 57% of the value ($8.2 \text{ m}^3 \text{ ha}^{-1}$) was attributable to positive effects, 65% ($9.3 \text{ m}^3 \text{ ha}^{-1}$) to competitive effects and -23% ($-3.2 \text{ m}^3 \text{ ha}^{-1}$) to negative effects, compared to 63% and 37% for CE and SE with additive partitioning (Figure 2c, d).

Experimental grassland mixtures

Mahaut et al. (2020) conducted an experimental study in pots with six grassland species in full and half density monocultures, as well as in every combination of two, three and six species in mixture. The studied species vary in productivity and functional traits. We used aboveground biomass data from full and half density monocultures, as well as all combinations of two species in full density mixtures (Table 1). Species competitive ability was approximated with species biomass in full density monocultures.

Competitive partitioning suggested a weak facilitative effect in two mixtures that were associated with *Lotus corniculatus*, one of the two legume species studied (Tables 1 and S2). Community negative effects occurred in two mixtures with *Sanguisorba minor* and between the two legumes, due to negative effects of *Sanguisorba minor* on other species and *Lotus corniculatus* on *Trifolium repens* (Table S2). Across all 15 two-species mixtures, the average of net biodiversity effects on aboveground biomass was 3.9 g/pot or 49% of the full density monoculture yield averaged across all species. On a relative basis, 60% (2.3 g) of the value was attributable to positive effects, 45% (1.7 g) to competitive effects, and -5% (-0.2 g) to negative effects, compared to 84% and 16% for CE and SE with additive partitioning (Table 2).

Comparison of additive and competitive partitioning

The competitive partitioning model provides a conceptual framework that helps establish the linkage between changes in ecosystem productivity and effects of species interactions. Under this framework, in both simulated tree and experimental grassland mixtures, net biodiversity effects were generally positive, resulting from both positive and competitive interactions. The positive competitive effects result from greater competitive yield responses of more competitive species (Huston, 1997; Montazeaud et al., 2018; Tilman et al., 1997; Wardle, 1999), i.e., the yield

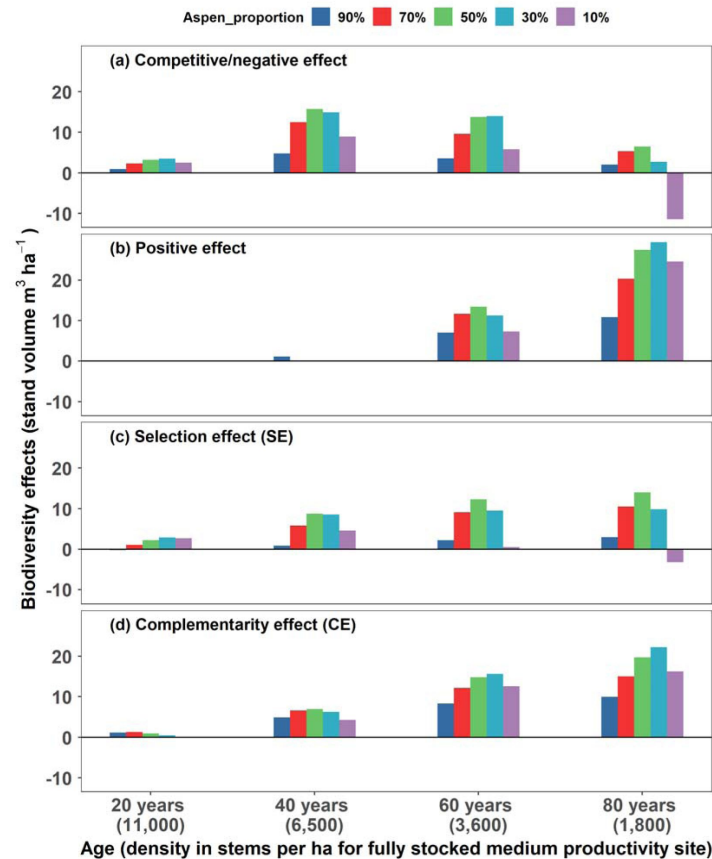


Figure 2.

Partitioning net biodiversity effects (changes in stand volume in mixtures relative to full density monocultures) with competitive (a and b) and additive (c and d) partitioning models for 20- to 80-year-old mixed trembling aspen and white spruce with varying species compositions from nearly pure aspen (90% aspen) to nearly pure spruce (90% spruce) on average medium productivity sites in western Canada. The growth and yield data (Table 1) are generated with GYPsy (Huang et al., 2009) and calculations of biodiversity components with this figure are detailed in Table S1.

Species mixture	Competitive partitioning		Additive partitioning	
0.50Species1: 0.50Species2	Positive effect	Competitive/ negative effect	Complementarity effect (CE)	Selection effect (SE)
Bromus:Dactylis	1.8	1.7	2.7	0.9
Bromus: Lotus	1.2	-0.1	1.1	-0.1
Bromus:Plantago	2.5	3.2	4.3	1.4
Bromus:Sanguisorba	0.1	0.1	0.1	0.0
Bromus:Trifolium	0.9	2.8	3.6	0.0
Dactylis:Lotus	4.5	1.2	4.2	1.6
Dactylis:Plantago	4.3	0.3	5.1	-0.4
Dactylis:Sanguisorba	0.0	0.8	0.3	0.5
Dactylis:Trifolium	3.0	0.7	4.0	-0.3
Lotus:Plantago	7.6	2.3	7.9	1.9
Lotus:Sanguisorba	0.0	-0.9	-0.6	-0.3
Lotus:Trifolium	0.0	1.8	0.8	1.1
Plantago:Sanguisorba	1.5	4.7	3.9	2.3
Plantago:Trifolium	5.0	1.2	7.8	-1.6
Sanguisorba:Trifolium	2.9	3.6	3.9	2.6
Average	2.3	Competitive: 1.7 Negative: -0.2	3.3	0.6

Species codes: See Table 1.

Table 2.

Partitioning net biodiversity effects (changes in aboveground biomass, g) with competitive and additive partitioning models using experimental data (Table 1 [↗](#)) from a grassland diversity–productivity study (Mahaut et al., 2020 [↗](#)). Calculations are detailed in Table S2.

gain of more competitive species from increases in resource availability exceeds the yield loss of less competitive species from decrease in resource availability, due to different competitive growth responses and full density monoculture yields (Tables S1 and S2). The greater competitive yield response of more competitive trembling aspen in young and middle-aged mixtures resulted in net biodiversity effects that were predominantly from competitive interactions (**Figure 2a**, [b](#); Table S1). A facilitative effect was rarely detected, only on spruce in nearly pure spruce mature mixture, possibly due to improved nutrient availability by aspen (Man and Lieffers, 1999 ; Peterson and Peterson, 1992) and in mixtures with legume *Lotus corniculatus* in grassland mixtures, likely due to improved nitrogen availability (Mahaut et al., 2020). The strong negative effect of mixing on aspen (i.e., observed yields < competitive expectations, see Table S1) in young and middle-aged mixtures suggests that trembling aspen originating naturally in high density does not benefit from mixing with white spruce. This is in contrast with white spruce where the deviations of observed yields from competitive expectations were generally positive, likely due to its shade tolerance and positive effects of aspen on nutrient availability (Man and Lieffers, 1999 ; Peterson and Peterson, 1992).

The net biodiversity effects reveal overall community yield changes from what would be expected from full density monoculture yields (Loreau and Hector, 2001 , 2019). However, the underlying mechanisms of species interactions for detected changes are uncertain, due to effects of competitive interactions that can be positive or negative. In both simulated mixed forests and experimental grassland mixtures, the partial density monoculture yield of the more competitive species exceeded the total expected yield of both species in nearly all mixtures, resulting in positive biodiversity effects when positive species interactions are not involved. Such positive biodiversity effects may help understand changes in ecosystem productivity detected through the null expectation (Carroll et al., 2011 ; Godoy et al., 2020) but do not support the general expectation that positive changes in ecosystem productivity are associated with positive species interactions (Barry et al., 2019 ; Drake, 2003 ; Hooper et al., 2005 ; Loreau, 2000) or crop mixing in agriculture and forestry where interests are positive effects (Forrester and Pretzsch, 2015 ; Man and Lieffers, 1999 ; Montazeaud et al., 2018). Thus, partitioning net biodiversity effects by effects of species interactions (i.e., positive, competitive, negative effects) allows for estimating relative importance of mechanisms of species interactions responsible for changes in ecosystem productivity. This enables ecologists to determine which communities are truly benefitting from diversity in the form of positive interactions and which result from interspecific differences in growth and competitive ability. In the aspen and spruce mixtures examined, community positive effects occurred largely in pre-mature and mature mixtures (>=60 years), even though net biodiversity effects and CE were positive at nearly all stages of stand development (**Figure 2**). In grassland mixtures, two mixtures (i.e., *Dactylis glomerata* - *Sanguisorba minor* and *Lotus corniculatus* - *Trifolium repens*) had positive net biodiversity effects that resulted entirely from competitive interactions (Tables 2 and S2). These positive net biodiversity effects may indicate evidence for positive changes in ecosystem productivity when these changes were actually driven by differential outcomes of interspecific competition.

We demonstrate with simulated and experimental species mixtures that competitive interaction is a major source of the net biodiversity effects and affects the additive components of both CE and SE. In the scenario of partial density monocultures emulating a simplified scenario of competitive exclusion where positive net biodiversity effects resulted entirely from competitive interactions, CE was generally negative in tree mixtures where relative yield changes were smaller with more competitive species (negative mean relative yield change) but positive in grassland mixtures where relative yield changes were greater with more competitive species (positive mean relative yield change) (Tables S1 and S2). In aspen-spruce mixtures, SE was either larger (age 80) or smaller (age 20, 40, and 60) than competitive effect. In the grassland communities, SE was similar to competitive effect in 3 of the 15 mixtures (*Bromus erectus* - *Lotus corniculatus*, *Dactylis glomerata* - *Lotus corniculatus*, and *Lotus corniculatus* - *Sanguisorba minor*) but substantially smaller in the remaining mixtures (**Table 2**). Even in the extreme cases where SE was nearly zero (*Bromus*

erectus - *Sanguisorba minor* and *Bromus erectus* - *Trifolium repens*), competitive effect was still relatively strong, suggesting that a common assumption to attribute positive net biodiversity effects to positive species interactions when SE is close to zero (Feng et al., 2022 [↗](#); Loreau and Hector, 2001 [↗](#)) is not justified. Competitive interactions amplify interspecies differences and therefore SE, while species interference on more competitive species (*Bromus erectus* - *Sanguisorba minor*) and greater positive species interactions on less competitive species (*Bromus erectus* - *Trifolium repens*) (Table S2) reduce SE. Our findings, along with some recent suggestions (Bourrat et al., 2023 [↗](#); Carroll et al., 2011 [↗](#); Godoy et al., 2020 [↗](#); Turnbull et al., 2013 [↗](#)), do not support some common exercises that attribute positive/negative CEs to positive/negative interactions (Barry et al., 2019 [↗](#); Fargione et al., 2007 [↗](#); Feng et al., 2022 [↗](#); Hagan et al., 2021 [↗](#); Loreau and Hector, 2001 [↗](#); Petchey, 2003 [↗](#)) and positive SEs to competitive interactions (Loreau, 2000 [↗](#); Loreau and Hector, 2001 [↗](#); Montazeaud et al., 2018 [↗](#)).

Several assumptions are adopted with the development of competitive partitioning model. First, we assumed that more productive species are more competitive, which was mostly true with the species mixtures examined in terms of species yields in mixture relative to their full density monoculture yields (Table 1 [↗](#)). There were exceptions with *Dactylis*:*Plantago* and *Plantago*:*Trifolium* in grassland mixtures where less productive species *Plantago* based on full density monoculture yields performed better in mixtures, which may suggest positive effects on *Plantago* and negative effects on the other species. However, species yields in partial versus full density monocultures indicate that these exceptions resulted from the stagnation of more productive *Plantago* in full (high) density monocultures, due to restricted rooting volumes and supplies of water and nutrients in pots (Chung et al., 2021 [↗](#)). The release of growth stagnation in partial density monocultures and mixtures affects not only the mechanisms of changes in ecosystem productivity assessed with competitive and additive partitioning models but also the magnitudes of the changes in ecosystem productivity detected through the null expectation (Loreau and Hector, 2001 [↗](#)). In this case, the use of partial density monoculture yields in assessment of species competitive ability are probably more appropriate, which would increase the values of competitive effects and reduce those of positive effects based on the competitive expectations. Partial density monocultures help detect competition-related yield changes, which would be otherwise unknown. Second, we used partial density monocultures to determine maximum competitive growth responses in a simplified competitive exclusion where more competitive species are not affected and grow like partial density monocultures, while less competitive species are competitively eliminated. We assumed a linear relationship between competitive growth responses and species relative competitive ability (Figure 1 [↗](#)), which should be a good approximation (Gaudet and Keddy, 1988 [↗](#)), but may not be the most appropriate, e.g., light availability and plant height (Brüllhardt et al., 2020 [↗](#)). Linear models can overestimate or underestimate competitive expectations of all species and affect individual species assessments but would have limited impacts on community level assessments, due to different directions of competitive growth responses among species.

Estimation of maximum competitive growth responses in mixtures of more than two species or in low densities may require additional efforts. In mixtures where more than one species are above average competitive ability, maximum increase in resource availability would be smaller, $\frac{1 - \sum RY_{Ej}}{RY_{Ei}} - 1$ (j = species more competitive than species i), instead of $\frac{1}{RY_{Ei}} - 1$. Maximum competitive growth increases determined in partial density monocultures may need to be adjusted accordingly. For less competitive species, complete elimination would not occur in mixtures of extremely low densities or small individual size where growth is largely independent of resource availability. Proportional growth reduction ($\frac{w_{if}}{w_{ip}} - 1$, opposite of proportional growth increase in partial density monocultures $\frac{w_{ip}}{w_{if}} - 1$) or partial elimination (by more competitive species) may be more appropriate for estimating maximum competitive growth reductions.

The competitive partitioning model incorporates effects of competitive interactions into the conventional null expectation and assists: (1) understanding the mechanisms of species interactions driving positive biodiversity-productivity relationships (Cardinale et al., 2012 [↗](#); Isbell et al., 2017 [↗](#)) by relative contributions of positive interactions from more species of diverse niche (Hooper et al., 2005 [↗](#); Tilman et al., 1997 [↗](#)) and competitive interactions from greater yield responses of more competitive and more productive species to changes in resource availability in mixture (Aarssen, 1997 [↗](#); Huston, 1997 [↗](#); Pillai and Gouhier, 2019 [↗](#); Tilman et al., 1997 [↗](#)), (2) examining the mechanisms of key species in influencing ecosystem productivity by roles of competitive interactions due to their higher productivity and competitive advantages (Loreau and Hector, 2001 [↗](#); Mahaut et al., 2020 [↗](#)), (3) meaningful comparisons of changes in ecosystem productivity (i.e., relative yield totals) across different ecosystems based on competitive expectations, and (4) redefining changes in ecosystem productivity such that positive changes result from positive interactions and negative changes from negative interactions (Drake, 2003 [↗](#); Hooper et al., 2005 [↗](#); Loreau, 2000 [↗](#)). Under this competitive partitioning framework, null expectations are either raised for more competitive species with increased resource availability or lowered for less competitive species with decreased resource availability in mixture relative to full density monocultures. The magnitudes of competitive growth responses are derived from species relative competitive ability in mixture and their maximum competitive growth responses determined from full and partial density monocultures under a simplified scenario of competitive exclusion. Current biodiversity experiments do not have partial density monocultures and therefore do not provide estimates of competitive growth responses that would occur in mixture due to interspecies differences in growth and competitive ability. However, density-size/yield relationship is one of the most extensively studied areas in ecology (Watkinson, 1980 [↗](#); Weiner and Freckleton, 2010 [↗](#)). Previous research can help determine maximum competitive growth responses through density-size/yield relationships (Huang et al., 2009 [↗](#)). This means that the competitive partitioning model can be used to investigate changes in ecosystem productivity by effects of species interactions in current biodiversity experiments established with replacement design (Fargione et al., 2007 [↗](#); Loreau and Hector, 2001 [↗](#)).

Conclusions

Competitive interactions are the major source of biodiversity effects and affect the additive components of both CE and SE. The interpretations of CEs and SEs with specific mechanisms of species interactions lack mathematical or ecological justifications. For example, attributing positive biodiversity effects solely to effects of positive interactions, as commonly seen in the literature (Barry et al., 2019 [↗](#); Fargione et al., 2007 [↗](#); Feng et al., 2022 [↗](#); Hooper et al., 2005 [↗](#); Loreau and Hector, 2001 [↗](#)), assumes that competitive interactions are productivity-neutral (SE close to zero) (Feng et al., 2022 [↗](#); Loreau and Hector, 2001 [↗](#)). This assumption is true if ecosystem productivity is defined by total resources pool (Tilman et al., 2014 [↗](#)) but generally not true if mixture productivity is defined by species monoculture yields as with the additive partitioning (Loreau and Hector, 2001 [↗](#)), due to the predominance of competitive interactions in interspecific relationships (Goldberg and Barton, 1992 [↗](#)) and positive effects of competitive interactions on ecosystem productivity based on the null expectation.

The competitive partitioning model is based on ecological theories of interspecific relationships, competitive ability, and competitive exclusion principle and experimental designs of both replacement and additive series. This new partitioning approach enables meaningful assessments of species interactions at both species and community levels and provides detailed insights into the mechanisms of species interactions that drive changes in ecosystem productivity. We believe that our framework, which is admittedly perfectible, is one promising avenue to determine effects of species interactions responsible for changes in ecosystem productivity, an approach that is long sought after in biodiversity-ecosystem productivity research (Loreau and Hector, 2012, 2019; Mahaut et al., 2020 [↗](#)).

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Author contributions

R.M. and H.Y. designed the research; J.T., S.H. and R.M. generated growth and yield data with the GYPsy; and J.T., C.A.N., E.B.S., S.H., R.M., H.Y., G.T.F., C.V., and J.Z. contributed to the interpretations of the results and to the writing of the paper.

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Reviewer #1 (Public review):

As a starting point, the authors discuss the so-called "additive partitioning" (AP) method proposed by Loreau & Hector in 2001. The AP is the result of a mathematical rearrangement of the definition of overyielding, written in terms of relative yields (RY) of species in mixtures relative to monocultures. One term, the so-called complementarity effect (CE), is proportional to the average RY deviations from the null expectations that plants of both species "do the same" in monocultures and mixtures. The other term, the selection effect (SE), captures how these RY deviations are related to monoculture productivity. Overall, CE measures whether relative biomass gains differ from zero when averaged across all community members, and SE, whether the "relative advantage" species have in the mixture, is related to their productivity. In extreme cases, when all species benefit, CE becomes positive. When large species have large relative productivity increases, SE becomes positive. This is intuitively compatible with the idea that niche complementarity mitigates competition ($CE > 0$), or that competitively superior species dominate mixtures and thereby driver overyielding ($SE > 0$).

However, it is very important to understand that CE and SE capture the "statistical structure" of RY that underlies overyielding. Specifically, CE and SE are not the ultimate biological mechanisms that drive overyielding, and never were meant to be. CE also does not describe niche complementarity. Interpreting CE and SE as directly quantifying niche complementarity or resource competition, is simply wrong, although it sometimes is done. The criticism of the AP method thus in large part seems unwarranted. The alternative methods the authors discuss (lines 108-123) are based on very similar principles.

The authors now set out to develop a method that aims at linking response patterns to "more true" biological mechanisms.

Assuming that "competitive dominance" is key to understanding mixture productivity, because "competitive interactions are the predominant type of interspecific relationships in plants", the authors introduce "partial density" monocultures, i.e. monocultures that have the same planting density for a species as in a mixture. The idea is that using these partial density monocultures as a reference would allow for isolating the effect of competition by the surrounding "species matrix".

The authors argue that "To separate effects of competitive interactions from those of other species interactions, we would need the hypothesis that constituent species share an identical niche but differ in growth and competitive ability (i.e., absence of positive/negative interactions)." - I think the term interaction is not correctly used here, because clearly competition is an interaction, but the point made here is that this would be a zero-sum game.

The authors use the ratio of productivity of partial density and full-density monocultures, divided by planting density, as a measure of "competitive growth response" (abbreviated as MG). This is the extra growth a plant individual produces when intraspecific competition is reduced.

Here, I see two issues: first, this rests on the assumption that there is only "one mode" of competition if two species use the same resources, which may not be true, because intraspecific and interspecific competition may differ. Of course, one can argue that then somehow "niches" are different, but such a niche definition would be very broad and go beyond the "resource set" perspective the authors adopt. Second, this value will heavily depend on timing and the relationship between maximum initial growth rates and competitive abilities at high stand densities.

The authors then progress to define relative competitive ability (RC), and this time simply uses monoculture biomass as a measure of competitive ability. To express this biomass in a

standardized way, they express it as different from the mean of the other species and then divide by the maximum monoculture biomass of all species.

I have two concerns here: first, if competitive ability is the capability of a species to preempt resources from a pool also accessed by another species, as the authors argued before, then this seems wrong because one would expect that a species can simply be more productive because it has a broader niche space that it exploits. This contradicts the very narrow perspective on competitive ability the authors have adopted. This also is difficult to reconcile with the idea that specialist species with a narrow niche would outcompete generalist species with a broad niche. Second, I am concerned by the mathematical form. Standardizing by the maximum makes the scaling dependent on a single value.

As a final step, the authors calculate a "competitive expectation" for a species' biomass in the mixture, by scaling deviations from the expected yield by the product $MG \times RC$. This would mean a species does better in a mixture when (1) it benefits most from a conspecific density reduction, and (2) has a relatively high biomass.

Put simply, the assumption would be that if a species is productive in monoculture (high RC), it effectively does not "see" the competitors and then grows like it would be the sole species in the community, i.e. like in the partial density monoculture.

Overall, I am not very convinced by the proposed method.

Comments on revised version:

Only minimal changes were made to the manuscript, and they do not address the main points that were raised.

<https://doi.org/10.7554/eLife.98073.2.sa3>

Reviewer #2 (Public review):

This manuscript by Tao et al. reports on an effort to better specify the underlying interactions driving the effects of biodiversity on productivity in biodiversity experiments. The authors are especially concerned with the potential for competitive interactions to drive positive biodiversity-ecosystem functioning relationships by driving down the biomass of subdominant species. The authors suggest a new partitioning schema that utilizes a suite of partial density treatments to capture so-called competitive ability. While I agree with the authors that understanding the underlying drivers of biodiversity-ecosystem functioning relationships is valuable - I am unsure of the added value of this specific approach for several reasons.

Comments on revised version:

The authors changed only one minor detail in response to the last round of reviews.

<https://doi.org/10.7554/eLife.98073.2.sa2>

Reviewer #3 (Public review):

Summary:

This manuscript claims to provide a new null hypothesis for testing the effects of biodiversity on ecosystem functioning. It reports that the strength of biodiversity effects changes when this different null hypothesis is used. This main result is rather inevitable. That is, one expects a different answer when using a different approach. The question then becomes

whether the manuscript's null hypothesis is both new and an improvement on the null hypothesis that has been in use in recent decades.

Strengths:

In general, I appreciate studies like this that question whether we have been doing it all wrong and I encourage consideration of new approaches.

Weaknesses:

Despite many sweeping critiques of previous studies and bold claims of novelty made throughout the manuscript, I was unable to find new insights. The manuscript fails to place the study in the context of the long history of literature on competition and biodiversity and ecosystem functioning. The Introduction claims the new approach will address deficiencies of previous approaches, but after reading further I see no evidence that it addresses the limitations of previous approaches noted in the Introduction. Furthermore, the manuscript does not reproducibly describe the methods used to produce the results (e.g., in Table 1) and relies on simulations, claiming experimental data are not available when many experiments have already tested these ideas and not found support for them. Finally, it is unclear to me whether rejecting the 'new' null hypothesis presented in the manuscript would be of interest to ecologists, agronomists, conservationists, or others.

Comments on revised version:

Please see review comments on the previous version of this manuscript. The authors have not revised their manuscript to address most of the issues previously raised by reviewers.

<https://doi.org/10.7554/eLife.98073.2.sa1>

Author response:

The following is the authors' response to the original reviews.

A summary of changes

- (1) Line 93: “positive effect” to “positive contribution”, as suggested by reviewer 2.
- (2) Line 147-148: the null hypothesis to test “equal interspecific and intraspecific interactions”, as indicated by reviewers 2 and 4.
- (3) Lines 155-162: removed to reduce duplication with the additive partitioning, as suggested by reviewer 2.
- (4) Lines 186-188: added “the estimated competitive growth response would also include the effects of density-dependent pests, pathogens, or microclimates”, as suggested by reviewer 3.
- (5) Lines 219-222: added “The community positive effect can be further partitioned by mechanisms of positive interactions (resource partitioning and facilitation), and facilitative effect can be classified as mutualism (+/+), commensalism (+/0), or parasitic (+/-) based on species specific assessments”.
- (6) Lines 377-386: added options for determining maximum competitive growth response in some extreme scenarios of species mixtures.
- (7) Figure 1: modified to show the variations of competitive growth response with relative competitive ability from minimum (null expectation) to maximum (competitive exclusion).

A summary of four reviewers' questions and authors' response

(1) A summary of authors' responses. Reviewers did not seem to understand our work. They indicated that our model is inadequate for hypothesis testing. The fact is, as we note below, that our model allows for more hypothesis testing than the additive partitioning model. They suggested that one of our model components, the competitive growth response, needs to be further partitioned. However, this term represents only the competition effect and can not be split any further. Reviewers criticized us for misunderstanding the additive components while they suggested the same logic to test some intuitive ideas. They did not seem to know that the effects of competitive interactions vary with assessment methods, which differ between competition and biodiversity research. Our work seeks to harmonise definitions between these two fields and bridge the gap. The reviewers acknowledged that the additive components (i.e., the selection effect and complementarity effect) do not have clear biological meanings; however, they did not acknowledge that the additive components are used extensively for determining mechanisms of species interactions in biodiversity research. There is hardly any research that uses the additive partitioning model without linking the additive components to specific mechanisms of species interactions (i.e., positive SE to competition and positive CE to positive interactions).

(2) Additive partitioning and underlying mechanisms. Some reviewers acknowledged that additive partitioning is not meant for determining mechanisms of species interactions and therefore argued that the additive partitioning should not be criticized for lack of biological meanings with the additive components. However, they insisted that additive partitioning is useful in quantifying net biodiversity effects against the null hypothesis that there is no difference between intraspecific and interspecific interactions or testing the idea that “niche complementarity mitigates competition” or “competitively superior species dominate mixtures”. Are these views contradictory each other? How can the additive partitioning that is not designed for determining mechanisms of species interactions provide meaningful explanations for outputs of species interactions, e.g., “niche complementarity mitigates competition” or “competitively superior species dominate mixtures”?

Reviewers did not seem to realize that these ideas are equivalent to the suggestions that CE represents for the effects of positive interactions and SE for the effects of competitive interactions, that the quantification of net biodiversity effects does not require the two additive components, and that the null hypothesis exists long before the additive partitioning (see de Wit, 1960, de Wit et al., 1966). It is generally agreed that CE and SE result from mathematical calculations and do not have clear biological meanings in terms of linkages to specific mechanisms of species interactions responsible for observed net biodiversity effects or changes in ecosystem function (Loreau and Hector, 2012; Bourrat et al., 2023). Calling some mixed effects of species interactions as mechanisms (e.g., CE and SE) is misleading.

Model structure: incomplete or inadequate for hypothesis testing. Other than positive, negative, and competition interactions, two reviewers wanted to have more specific interactions such as microclimate amelioration and negative feedback from species-specific pests and pathogens. The determination of these specific mechanisms requires more investigations and cannot be simply made through partitioning growth and yield data. However, the effects of these interactions will be captured in our definition of species interactions. Reviewers did not seem to know that the additive partitioning would also not allow identifying these specific positive species interactions.

Inspired by the mathematical form of additive partitioning, two reviewers suggested that our model (presumably equation 4) is incomplete and the second term, i.e., competitive growth response needs to be further explored or partitioned. The second term represents deviations from the null expectation, due to species differences in growth and competitive ability or competition effect. We do not know why and how this term can be further partitioned and what any subcomponents would mean.

Our competitive partitioning model is based on two hypotheses: first, the null hypothesis to test the equivalence of interspecific and intraspecific interactions. This hypothesis is the same as the additive partitioning model. Second, the competitive hypothesis, which tests the dominance of positive or negative species interactions in a community. Thus, our model allows for more hypothesis testing than the current additive partitioning model.

(3) Types of species interactions. We follow the definition of species interactions generally used in biodiversity research (see Loreau and Hector, 2001), i.e., positive interactions (or complementarity) include resource partitioning and facilitation, negative interactions include interference competition, and competitive interactions include resource competition. One reviewer suggested that resource partitioning is byproduct of competition and should not be part of positive species interactions, which may be true for long-term evolution of species co-existence but not for biodiversity experiments of decade duration at most. Two reviewers suggested that positive interactions should also include microclimate amelioration or negative feedback from species-specific pests and pathogens. We agree and these are included in our definition.

(4) Significance of partial density monocultures. We used partial and full density monocultures and species competitive ability to determine what species can possibly achieve in mixture under the competitive hypothesis that constituent species share an identical niche but differ in growth and competitive ability. We did not use partial monocultures to test the effects of density on biodiversity effects. As with the additive partitioning, the competitive partitioning model is not designed for comparing yields across different densities. We added at lines 186-188 to indicate that the estimated competitive growth response would also include the effects of density-dependent pests, pathogens, or microclimates.

Similarly, we do not use the partial density monoculture to supplant the replacement series design. Partial density monocultures only supplement the “replacement series” design that does not provides estimates of facilitative effects and competitive growth responses that would occur in mixtures. It is crucial to know that one experimental approach is simply not enough for determining underlying mechanisms of species interactions responsible for changes in ecosystem function.

(5) Competition effect in competition and biodiversity research. Due to different methods used, competition effect in competition research has different ecological meanings from that in biodiversity research. In competition research, species performance in mixture are compared with their partial density monocultures and therefore competition effect is generally negative, as suggested by reviewer 4. In biodiversity research, comparison is between mixture and full density monocultures. The resulting competition effect can be positive or negative for both individual species and community productivity defined by species composition and full density monoculture yields.

Therefore, we cannot use the results of competition research based on additive series design to describe effects of competitive interactions on ecosystem productivity based replacement series design.

Reviewer #1 (Public Review):

[Editors' note: this is an overall synthesis from the Reviewing Editor in consultation with the reviewers.]

The three reviews expand our critique of this manuscript in some depth and complementary directions. These can be synthesized in the following main points (we point out that there is quite a bit more that could be written about the flaws with this study; however, time constraints prevented us from further elaborating on the issues we see):

(1) *It is unclear what the authors want to do.*

As indicated by the title, our objective is to “partition changes in ecosystem productivity by effects of species interactions”, i.e., partitioning net biodiversity effects estimated from the null expectation into components associated with positive, negative, or competition interspecific interactions.

It seems their main point is that the large BEF literature and especially biodiversity experiments overstate the occurrence of positive biodiversity effects because some of these can result from competition.

We demonstrated through ecological theories and simulation/experiment data that competition is a major source of the net biodiversity effects estimated with additive partitioning model. We know that competition effect varies with mixture attributes. Future research will determine average effect of competitive interactions on biodiversity effects in large BEF literature.

Because reduced interspecific relative to intraspecific competition in mixture is sufficient to produce positive effects in mixtures (if interspecific competition = 0 then $RYT = S$, where S is species richness in mixture -- this according to the reciprocal yield law = law of constant final yield), they have a problem accepting $NE > 0$ as true biodiversity effect (see additive partitioning method of Loreau & Hector 2001 cited in manuscript).

We have no problem to accept $NE > 0$ as true positive biodiversity effect. However, $NE > 0$ can also result from competitive interactions based on the null expectation and needs to be partitioned by effects of species interactions.

(2) *The authors' next claim, without justification, that additive partitioning of NE is flawed and theoretically and biologically meaningless.*

The additive partitioning model is based on Covariance equation (or Price equation) that has nothing to do with biodiversity partitioning (Bourrat et al., 2023). Biological meaning was arbitrarily assigned to CE and SE. We made clear that the additive partitioning model is mathematically sound but does not have biological meanings that it has been used for.

They misinterpret the CE component as biological niche partitioning and the SE component as biological dominance.

We did not. Loreau and Hector (2001) clearly indicated positive CE for positive interactions and positive SE for competitive interactions, which is generally what has been used for in the last twenty years.

They do not seem to accept that the additive partitioning is a logically and mathematically sound derivation from basic principles that cannot be contested.

We do not have problem with mathematical form of additive partitioning but only oppose ecological meanings assigned to CE and SE, simply because CE and SE both result from all species interactions (see Loreau and Hector, 2001; Bourrat et al., 2023). The reviewer seemed to have a contradictory thinking that the additive components are biologically meaningless but derived from biological basic principles.

(3) *The authors go on to introduce a method to calculate species-level overyielding ($RY > 1/S$ in replacement series experiments) as a competitive growth response and multiply*

this with the species monoculture biomass relative to the maximum to obtain competitive expectation. This method is based on resource competition and the idea that resource uptake is fully converted into biomass (instead of e.g. investing it in allelopathic chemical production).

Correct, but we did not assume “resource uptake is fully converted into biomass”.

(4) It is unclear which experiments should be done, i.e. are partial-density monocultures planted or simply calculated from full-density monocultures? At what time are monocultures evaluated? The framework suggests that monocultures must have the full potential to develop, but in experiments, they are often performing very poorly, at least after some time. I assume in such cases the monocultures could not be used.

Both partial and full density monocultures are needed, along with mixtures to separate NE by species interactions. Calculating competitive growth responses from density-size relationships can be an alternative, given the lack of partial density monocultures in current biodiversity experiments, but is not preferred.

Similar to additive partitioning, our model can (and should) be applied to all developmental stages of an experiment to examine how interactions evolve through time.

(5) There are many reasons why the ideal case of only resource competition playing a role is unrealistic. This excludes enemies but also differential conversion factors of resources into biomass and antagonistic or facilitative effects. Because there are so many potential reasons for deviations from the null model of only resource competition, a deviation from the null model does not allow conclusions about underlying mechanisms.

The competitive expectation is only a hypothesis, just as the null expectation. The difference between competitive and null expectations represents a competitive effect resulting from species differences in growth and competitive ability, while the deviation of observed yields from the competitive expectation indicates positive or negative effect (see lines 201-219).

Furthermore, this is not a systematically developed partitioning, but some rather empirical ad hoc formulation of a first term that is thought to approximate competitive effects as understood by the authors (but again, there already are problems here). The second residual term is not investigated. For a proper partitioning approach, one would have to decompose overyielding into two (or more) terms and demonstrate (algebraically) that under some reasonable definitions of competitive and non-competitive interactions, these end up driving the respective terms.

The first term represents the null expectation assuming equal interspecific and intraspecific interactions, i.e., absence of positive, negative, and competition effects. The second residual term represents competition effect, due to species differences in growth and competitive ability. The meaning of second residual term is clear and does not need to be further partitioned or investigated.

In fact, our competitive partitioning also has several components including null expectation, competitive growth response, and observed yield, plus partial density monocultures for species assessment, or null expectations, competitive expectations, and observed yields for community level assessment, although different from the additive partitioning.

(6) Using a simplistic simulation to test the method is insufficient. For example, I do not see how the simulation includes a mechanism that could create CE in additive

partitioning if all species would have the same monoculture yield. Similarly, they do not include mechanisms of enemies or antagonistic interactions (e.g. allelopathy).

The simulation model we used is developed from real world data and can only do what are available in the model in terms of species and their growth under different conditions. We can not go beyond data limitation. The model is empirical and has been shown to accurately estimate yield in the aspen-spruce forest condition. We would also note that we do also use experimental data (Table 2).

(7) The authors do not cite relevant literature regarding density x biodiversity experiments, competition experiments, replacement-series experiments, density-yield experiments, additive partitioning, facilitation, and so on.

We cited literature relevant to biodiversity partitioning since we are not aiming to cover everything. The reviewer may not be aware that most of the research areas listed are actually included in our work, such as additive and replacement-series experiment designs, additive partitioning, facilitation, competition studies, and density-yield relationships. Our competitive model partitioning is based on biological principles, while the additive partitioning model is based only on a mathematical equation.

Overall, this manuscript does not lead further from what we have already elaborated in the broad field of BEF and competition studies and rather blurs our understanding of the topic.

The results of competition studies based on additive series design are not really used in the broad field of BEF based on replacement series design. The effects of competitive interactions on BEF are never clearly defined using the results of competition studies. Our work is filling that gap.

Reviewer #2 (Public Review):

This manuscript is motivated by the question of what mechanisms cause overyielding in mixed-species communities relative to the corresponding monocultures. This is an important and timely question, given that the ultimate biological reasons for such biodiversity effects are not fully understood.

As a starting point, the authors discuss the so-called "additive partitioning" (AP) method proposed by Loreau & Hector in 2001. The AP is the result of a mathematical rearrangement of the definition of overyielding, written in terms of relative yields (RY) of species in mixtures relative to monocultures. One term, the so-called complementarity effect (CE), is proportional to the average RY deviations from the null expectations that plants of both species "do the same" in monocultures and mixtures. The other term, the selection effect (SE), captures how these RY deviations are related to monoculture productivity. Overall, CE measures whether relative biomass gains differ from zero when averaged across all community members, and SE, whether the "relative advantage" species have in the mixture, is related to their productivity. In extreme cases, when all species benefit, CE becomes positive. When large species have large relative productivity increases, SE becomes positive. This is intuitively compatible with the idea that niche complementarity mitigates competition (CE>0), or that competitively superior species dominate mixtures and thereby driver overyielding (SE>0).

The reviewer needs to know that these ideas are based on the same logic that positive CE represents the effects of positive interactions and positive SE represents the effects of competitive interactions. CE>0 or SE>0 can result from many different scenarios of species interactions, not necessarily "niche complementarity mitigates competition" or

“competitively superior species dominate mixtures”. $CE > 0$ and $SE > 0$ can occur alone or together. We simply can not tell underlying mechanisms of overyielding from mathematical calculations (CE and SE), as suggested by this reviewer later.

The reviewer criticizes us while using the same logic themselves.

However, it is very important to understand that CE and SE capture the "statistical structure" of RY that underlies overyielding. Specifically, CE and SE are not the ultimate biological mechanisms that drive overyielding, and never were meant to be. CE also does not describe niche complementarity. Interpreting CE and SE as directly quantifying niche complementarity or resource competition, is simply wrong, although it sometimes is done. The criticism of the AP method thus in large part seems unwarranted. The alternative methods the authors discuss (lines 108-123) are based on very similar principles.

The reviewer actually supports our point. However, CE and SE have been largely used as biological mechanisms, positive CE as the results of complementary interactions and positive SE as the results of competitive interactions (see Loreau and Hector, 2001).

We do not have problem with the "statistical structure" of AP; it is simply a covariance equation. It is important to know that CE and SE do not provide additional information on overyielding than NE in terms of underlying mechanisms of species interactions. Any attempt to investigate mechanism of overyielding with CE or SE can easily go wrong.

Our competitive partitioning model incorporates effects of competitive interactions into the conventional null expectation and allows for separating different effects of species interactions. In comparison, the additive partitioning model does not have this capacity, not even designed for this purpose, as suggested by this and other reviewers.

The authors now set out to develop a method that aims at linking response patterns to "more true" biological mechanisms.

Assuming that "competitive dominance" is key to understanding mixture productivity, because "competitive interactions are the predominant type of interspecific relationships in plants", the authors introduce "partial density" monocultures, i.e. monocultures that have the same planting density for a species as in a mixture. The idea is that using these partial density monocultures as a reference would allow for isolating the effect of competition by the surrounding "species matrix".

Correct.

The authors argue that "To separate effects of competitive interactions from those of other species interactions, we would need the hypothesis that constituent species share an identical niche but differ in growth and competitive ability (i.e., absence of positive/negative interactions)." - I think the term interaction is not correctly used here, because clearly competition is an interaction, but the point made here is that this would be a zero-sum game.

We did not say that competition is not an interaction; we only want to separate the effect of competition from those of other species interactions.

The authors use the ratio of productivity of partial density and full-density monocultures, divided by planting density, as a measure of "competitive growth response" (abbreviated as MG). This is the extra growth a plant individual produces when intraspecific competition is reduced.

Correct.

We added at lines 377-386 to discuss options to determine MG in some uncommon scenarios of species mixtures.

Here, I see two issues: first, this rests on the assumption that there is only "one mode" of competition if two species use the same resources, which may not be true, because intraspecific and interspecific competition may differ. Of course, one can argue that then somehow "niches" are different, but such a niche definition would be very broad and go beyond the "resource set" perspective the authors adopt. Second, this value will heavily depend on timing and the relationship between maximum initial growth rates and competitive abilities at high stand densities.

First, the "competitive effect" focusses on resource competition and other forms of competition (presumably interference competition) are included in the negative interactions.

Second, competitive growth response varies over time and with density, and so do NE, CE, SE, and interspecific interactions.

The authors then progress to define relative competitive ability (RC), and this time simply uses monoculture biomass as a measure of competitive ability. To express this biomass in a standardized way, they express it as different from the mean of the other species and then divide by the maximum monoculture biomass of all species.

I have two concerns here: first, if competitive ability is the capability of a species to preempt resources from a pool also accessed by another species, as the authors argued before, then this seems wrong because one would expect that a species can simply be more productive because it has a broader niche space that it exploits. This contradicts the very narrow perspective on competitive ability the authors have adopted. This also is difficult to reconcile with the idea that specialist species with a narrow niche would outcompete generalist species with a broad niche. Second, I am concerned by the mathematical form. Standardizing by the maximum makes the scaling dependent on a single value.

First, growth conditions are controlled in biodiversity experiments, i.e., both monocultures and mixtures are the same in resource space. Species do not have opportunity to exploit resources outside experimental area. For example, if less productive species on normal soils outperform more competitive species on saline/alkaline soil, these "less productive species" are considered "more productive".

Second, as discussed in our paper (lines 367-376; Figure 1), more research is needed to determine relationships between species traits (biomass or height) and relative competitive ability. By then, scaling by the maximum would not be needed. There has been quite a lot of research on such relationships; we should leave this to subject experts to determine what would be mostly appropriate for species studied.

As a final step, the authors calculate a "competitive expectation" for a species' biomass in the mixture, by scaling deviations from the expected yield by the product $MG \times RC$. This would mean a species does better in a mixture when (1) it benefits most from a conspecific density reduction, and (2) has a relatively high biomass.

Put simply, the assumption would be that if a species is productive in monoculture (high RC), it effectively does not "see" the competitors and then grows like it would be the sole species in the community, i.e. like in the partial density monoculture.

Correct, if species competitive ability differs substantially, the more competitive species in the mixture would grow like partial density monoculture. This extra growth should not be treated as sources of positive biodiversity effects, simply because it does not result from positive species interactions.

Overall, I am not very convinced by the proposed method.

(1) The proposed method seems not very systematic but rather "ad hoc". It also is much less a partitioning method than the AP method because the other term is simply the difference. It would be good if the authors investigated the mathematical form of this remainder and explored its properties.. when does complementarity occur? Would it capture complementarity and facilitation?

AP is, by no means, systematic. Remember, AP is based on covariance equation (or Price equation) that has nothing to do with species interactions, other than nice-looking mathematical form (Bourrat et al., 2023). Ecological meanings are subjectively given to CE and SE. Therefore, CE and SE reflect what we call them, not what they really mean.

The remainder measures deviations from the null expectation, due to only competition effect, and can not be partitioned any further. The remainder would be positive for more competitive species and negative for less competitive species in mixture relative to their full density monoculture. The deviation of observed yields from competitive expectations indicates dominance of positive or negative species interactions. All these are clearly outlined at lines 201-221.

(2) The justification for the calculation of MG and RC does not seem to follow the very strict assumptions of what competition (in the absence of complementarity) is. See my specific comments above.

We do not see why not.

(3) Overall, the manuscript is hard to read. This is in part a problem of terminology and presentation, and it would be good to use more systematic terms for "response patterns" and "biological mechanisms".

To help understand the variations of competitive growth response with relative competitive ability, the x axis of Figure 1 is labelled with null expectation, competitive expectation, and competitive exclusion from minimum to maximum deviation of competitive ability from community average.

We have followed terms used in biodiversity partitioning and changing terms can be confusing.

Examples:

- on line 30, the authors write that CE is used to measure "positive" interactions and SE to measure "competitive interactions", and later name "positive" and "negative" interactions "mechanisms of species interactions". Here the authors first use "positive interaction" as any type of effect that results in a community-level biomass gain, but then they use "interaction" with reference to specific biological mechanisms (e.g. one species might attract a parasite that infests another species, which in turn may cause further changes that modify the growth of the first and other species).

There are some differences in meaning, but that is what CE and SE have been generally used for. Using different terms can be confusing and does not help understanding the problems

with AP.

- on line 70, the authors state that "positive interaction" increases productivity relative to the null expectation, but it is clear that an interaction can have "negative" consequences for one interaction partner and "positive" ones for the other. Therefore, "positive" and "negative" interactions, when defined in this way, cannot be directly linked to "resource partitioning" and "facilitation", and "species interference" as the authors do. Also, these categories of mechanisms are still simple. For example, how do biotic interactions with enemies classify, see above?

We are explaining effects of competitive interactions on species yield, and ultimately on community yield that can be linked to “resource partitioning” and “facilitation”, and “species interference”.

More specific species interactions require detailed biological investigation and cannot be determined through partitioning of biomass production.

- line 145: "Under the null hypothesis, species in the mixture are assumed to be competitively equivalent (i.e., absence of interspecific interactions)". This is wrong. The assumption is that there are interspecific interactions, but that these are the same as the intraspecific ones. Weirdly, what follows is a description of the AP method, which does not belong here. This paragraph would better be moved to the introduction where the AP method is mentioned. Or omitted, since it is basically a repetition of the original Loreau & Hector paper.

As suggested, “absence of interspecific interactions” was replaced with “equal interspecific and intraspecific interactions”.

We have removed lines 155-162 to reduce duplication. However, our method is based on null expectation that needs to be introduced, despite it is part of AP.

Other points:

- line 66: community productivity, not ecosystem productivity.

Both community productivity and ecosystem productivity are used in biodiversity research, although meaning can be slightly different. Comparatively, ecosystem productivity is more common.

- line 68: community average responses are with respect to relative yields - this is important!

- line 64: what are "species effects of species interactions"?

We searched and did not find “species effects of species interactions”.

- line 90: here "competitive" and "productive" are mixed up, and it is important to state that "suffers more" refers to relative changes, not yield changes.

It, in fact, refers to yield changes. For example, less productive species, at active growth, are more responsive to changes in competition, while more productive species, at inactive growth (i.e., aging), are less responsive to changes in competition.

- line 92: "positive effect of competitive dominance": I don't understand what is meant here.

The phrase was modified to “positive contribution of competitive dominance to ecosystem productivity based on the null expectation”.

Reviewer #3 (Public Review):

Summary:

This manuscript by Tao et al. reports on an effort to better specify the underlying interactions driving the effects of biodiversity on productivity in biodiversity experiments. The authors are especially concerned with the potential for competitive interactions to drive positive biodiversity-ecosystem functioning relationships by driving down the biomass of subdominant species. The authors suggest a new partitioning schema that utilizes a suite of partial density treatments to capture so-called competitive ability. While I agree with the authors that understanding the underlying drivers of biodiversity-ecosystem functioning relationships is valuable - I am unsure of the added value of this specific approach for several reasons.

Strengths:

I can find a lot of value in endeavouring to improve our understanding of how biodiversity-ecosystem functioning relationships arise. I agree with the authors that competition is not well integrated into the complementarity and selection effect and interrogating this is important.

Weaknesses:

(1) The authors start the introduction very narrowly and do not make clear why it is so important to understand the underlying mechanisms driving biodiversity-ecosystem functioning relationships until the end of the discussion.

There are different ways to start introduction; we believe that starting with the problems of the current approach is the most effective for outlining the study's objective.

(2) The authors criticize the existing framework for only incorporating positive interactions but this is an oversimplification of the existing framework in several ways:

We did not criticize the existing framework for only incorporating positive interactions. We criticize the existing framework, because it is not based on mechanisms of species interactions, but is extensively used to determine underlying mechanisms driving biodiversity-ecosystem functioning relationships.

a. The existing partitioning scheme incorporates resource partitioning which is an effect of competition.

Resource partitioning means that species utilize resources differently, while competition means species use the same resources. “resource partitioning is an effect of competition” is not true in biodiversity experiments that are often short in duration and controlled in conditions.

b. The authors neglect the potential that negative feedback from species-specific pests and pathogens can also drive positive BEF and complementarity effects but is not a positive interaction, necessarily. This is discussed in Schnitzer et al. 2011, Maron et al. 2011, Hendriks et al. 2013, Barry et al. 2019, etc.

We did not. The feedback effect will be reflected in the differences between observed yields and competitive expectations if species in mixtures have different pests and pathogens relative to monocultures. The additive partitioning does not identify these feedback effects either.

c. Hector and Loreau (and many of the other citations listed) do not limit competition to SE because resource partitioning is a byproduct of competition.

Positive SE has been largely interpreted as the result of competition including Hector and Loreau (2001) and many others. It needs to be clear that neither of the additive components can be linked to specific mechanisms of species interactions.

Does “resource partitioning is a byproduct of competition” mean that species change their niche to avoid competition? If this is what the reviewer means, it may occur through long-term evolution, but not in short-term biodiversity experiments. Hector and Loreau (2001) clearly indicated that their complementarity effect includes both resource partitioning and facilitation.

(3) It is unclear how this new measure relates to the selection effect, in particular. I would suggest that the authors add a conceptual figure that shows some scenarios in which this metric would give a different answer than the traditional additive partition. The example that the authors use where a dominant species increases in biomass and the amount that it increases in biomass is greater than the amount of loss from it outcompeting a subdominant species is a general example often used for a selection effect when exactly would you see a difference between the two?:

a. Just a note - I do think you should see a difference between the two if the species suffers from strong intraspecific competition and has therefore low monoculture biomass but this would tend to also be a very low-density monoculture in practice so there would potentially be little difference between a low density and high-density monoculture because the individuals in a high-density monoculture would die anyway. So I am not sure that in practice you would really see this difference even if partial density plots were incorporated.

Linking new measure to SE or CE would be difficult (see many comparisons in Tables and Figures in our manuscript), as SE and CE are derived from mathematical equation and do not represent specific mechanisms of species interactions (Hector and Loreau 2012; Bourrat et al., 2023).

(4) One of the tricky things about these endeavors is that they often pull on theory from two different subfields and use similar terminology to refer to different things. For example - in competition theory, facilitation often refers to a positive relative interaction index (this seems to be how the authors are interpreting this) while in the BEF world facilitation often refers to a set of concrete physical mechanisms like microclimate amelioration. The truth is that both of these subfields use net effects. The relative interaction index is also a net outcome as is the complementarity effect even if it is only a piece of the net biodiversity effect. Trying to combine these two subfields to come up with a new partitioning mechanism requires interrogating the underlying assumptions of both subfields which I do not see in this paper.

Agree, microclimate amelioration is also part of positive effect and will be reflected in the difference between observed yield and competitive expectation. We can not separate the two mechanisms of positive species interactions without investigating influences of microclimate on growth and yield.

(5) The partial density treatment does not isolate competition in the way that the authors indicate. All of the interactions that the authors discuss are density-dependent including the mechanism that is not discussed (negative feedback from species-specific pests and pathogens). These partial density treatment effects therefore cannot simply be equated to competition as the authors indicate.:

We use partial density monoculture to determine maximum competitive growth response, effect of density-dependent intraspecific interactions, and species competitive ability to determine the level of maximum competitive growth response species can achieve in mixtures. There may be changes in species-specific pests and pathogens from partial to full density monocultures, which will be captured in competitive growth responses of individuals. We added at lines 186-188 to indicate that the maximum competitive growth response estimated would also include the effects of density-dependent pests, pathogens, or microclimates.

a. Additionally - the authors use mixture biomass as a stand-in for competitive ability in some cases but mixture biomass could also be determined by the degree to which a plant is facilitated in the mixture (for example).

We used monoculture biomass, not mixture biomass, to assess competitive ability

(6) I found the literature citation to be a bit loose. For example, the authors state that the additive partition is used to separate positive interactions from competition (lines 70-76) and cite many papers but several of these (e.g. Barry et al. 2019) explicitly do not say this.

Barry et al. (2019) defined CE as overproduction from monocultures, an effect of positive interactions.

(7) The natural take-home message from this study is that it would be valuable for biodiversity experiments to include partial density treatments but I have a hard time seeing this as a valuable addition to the field for two reasons:

a. In practice - adding in partial density treatments would not be feasible for the vast majority of experiments which are already often unfeasibly large to maintain.

The reviewer suggested that quantity is more important than quality. Without partial density monocultures no one can separate different effects of species interactions, as suggested by Loreau and Hector, reviewers, and many others that effects of species interactions can not be clearly differentiated with replacement series design. Unreliable scientific findings are not valuable.

b. The density effect would likely only be valuable during the establishment phase of the experiment because species that are strongly limited by intraspecific competition will die in the full-density plots resulting in low-density monocultures. You can see this in many biodiversity experiments after the first years. Even though they are seeded (or rarely planted) at a certain density, the density after several years in many monocultures is quite low.

True. High or low density also depends on individual size; if individuals do not get enough resources, density is high. Therefore, density effect can be strong even as density drops substantially from initial levels.

Reviewer #4 (Public Review):

Summary:

This manuscript claims to provide a new null hypothesis for testing the effects of biodiversity on ecosystem functioning. It reports that the strength of biodiversity effects changes when this different null hypothesis is used. This main result is rather inevitable. That is, one expects a different answer when using a different approach. The question then becomes whether the manuscript's null hypothesis is both new and an improvement on the null hypothesis that has been in use in recent decades.

It needs to be clear that we use two hypotheses, null hypothesis that is currently used with AP, and competitive hypothesis that is new with this manuscript. The null hypothesis helps determine changes in ecosystem productivity from all species interactions, while the competitive hypothesis helps partition changes in ecosystem productivity by mechanisms of species interactions, i.e., positive, negative, or competitive interactions.

Strengths:

In general, I appreciate studies like this that question whether we have been doing it all wrong and I encourage consideration of new approaches.

Weaknesses:

Despite many sweeping critiques of previous studies and bold claims of novelty made throughout the manuscript, I was unable to find new insights. The manuscript fails to place the study in the context of the long history of literature on competition and biodiversity and ecosystem functioning. The Introduction claims the new approach will address deficiencies of previous approaches, but after reading further I see no evidence that it addresses the limitations of previous approaches noted in the Introduction. Furthermore, the manuscript does not reproducibly describe the methods used to produce the results (e.g., in Table 1) and relies on simulations, claiming experimental data are not available when many experiments have already tested these ideas and not found support for them. Finally, it is unclear to me whether rejecting the 'new' null hypothesis presented in the manuscript would be of interest to ecologists, agronomists, conservationists, or others. I will elaborate on each of these points below.

First, there are many biodiversity experiments but those with partial density monocultures are rare. We found only one greenhouse experiment. We have to use simulation to illustrate different scenarios of species interactions to demonstrate how our approach works and how different it is from the AP.

Because of different methods used, the results of long history competition research (generally based on additive series design) cannot be used to define effects of competitive interactions in biodiversity research (generally based on replacement series design). This may be the reason that few competition researchers were cited in Loreau and Hector (2001).

Our approach requires two hypotheses, null and competitive, and the meaning of deviation from these hypotheses are outlined at lines 201-221 for both individual species and community level assessments. Distinguishing changes in ecosystem productivity by species interactions would be of great interest to “ecologists, agronomists, conservationists, or others”.

The critiques of biodiversity experiments and existing additive partitioning methods are overstated, as is the extent to which this new approach addresses its limitations. For example, the critique that current biodiversity experiments cannot reveal the effects of

species interactions (e.g., lines 37-39) isn't generally true, but it could be true if stated more specifically. That is, this statement is incorrect as written because comparisons of mixtures, where there are interspecific and intraspecific interactions, with monocultures, where there are only intraspecific interactions, certainly provide information about the effects of species interactions (interspecific interactions). These biodiversity experiments and existing additive partitioning approaches have limits, of course, for identifying the specific types of interactions (e.g., whether mediated by exploitative resource competition, apparent competition, or other types of interactions). However, the approach proposed in this manuscript gets no closer to identifying these specific mechanisms of species interactions. It has no ability to distinguish between resource and apparent competition, for example. Thus, the motivation and framing of the manuscript do not match what it provides. I believe the entire Introduction would need to be rewritten to clarify what gap in knowledge this proposed approach is addressing and what would be gained by filling this knowledge gap.

Our approach helps determine underlying mechanisms of species interactions, i.e., positive (resources partitioning or facilitation), negative, or competitive interactions. I am not sure how much we need to go further in identifying more specific mechanisms. If resource and apparent competition refers to resource and interference competition, our approach can tease apart them.

I recommend that the Introduction instead clarify how this study builds on and goes beyond many decades of literature considering how competition and biodiversity effects depend on density. This large literature is insufficiently addressed in this manuscript. This fails to give credit to previous studies considering these ideas and makes it unclear how this manuscript goes beyond the many previous related studies. For example, see papers and books written by de Wit, Harper, Vandermeer, Connolly, Schmid, and many others. Also, note that many biodiversity experiments have crossed diversity treatments with a density treatment and found no significant effects of density or interactions between density and diversity (e.g., Finn et al. 2013 Journal of Applied Ecology). Thus, claiming that these considerations of density are novel, without giving credit to the enormous number of previous studies considering this, is insufficient.

A misunderstanding here. Our approach is not designed to test density effect. The same density is held across full density monocultures and mixtures. We use partial density monocultures to determine what species may competitively achieve in full density mixture, without positive or negative interspecific interactions.

Replacement series designs emerged as a consensus for biodiversity experiments because they directly test a relevant null hypothesis. This is not to say that there are no other interesting null hypotheses or study designs, but one must acknowledge that many designs and analyses of biodiversity experiments have already been considered. For example, Schmid et al. reviewed these designs and analyses two decades ago (2002, chapter 6 in Loreau et al. 2002 OUP book) and the overwhelming consensus in recent decades has been to use a replacement series and test the corresponding null hypothesis.

Some wrong impressions. We are not trying to supplant “replacement series” with “additive series”; we use “additive series” designs to supplement “replacement series” design for partitioning changes in ecosystem productivity by mechanisms of species interactions, which would not be possible with “replacement series” design alone, as suggested by many including reviewers.

It is unclear to me whether rejecting the 'new' null hypothesis presented in the manuscript would be of interest to ecologists, agronomists, conservationists, or others. Most biodiversity experiments and additive partitions have tested and quantified diversity effects against the null hypothesis that there is no difference between intraspecific and interspecific interactions. If there was no less competition and no more facilitation in mixtures than in monocultures, then there would be no positive diversity effects. Rejecting this null hypothesis is relevant when considering coexistence in ecology, overyielding in agronomy, and the consequences of biodiversity loss in conservation (e.g., Vandermeer 1981 Bioscience, Loreau 2010 Princeton Monograph). This manuscript proposes a different null hypothesis and it is not yet clear to me how it would be relevant to any of these ongoing discussions of changes in biodiversity.

Our method begins with the null expectation: that intraspecific and interspecific interactions are equivalent. We then propose the competitive hypothesis as a second non-exclusive hypothesis which tests the dominance of positive or negative specific interactions. As shown by its name, the additive partitioning model has been advocated for partitioning biodiversity effects by some ecological mechanisms (CE and SE). The ecological meaning of deviation from the two hypotheses are outlined at lines 201-221 for both individual species and community level assessments.

The claim that all previous methods 'are not capable of quantifying changes in ecosystem productivity by species interactions and species or community level' is incorrect. As noted above, all approaches that compare mixtures, where there are interspecific interactions, to monocultures, where there are no species interactions, do this to some extent. By overstating the limitations of previous approaches, the manuscript fails to clearly identify what unique contribution it is offering, and how this builds on and goes beyond previous work.

The reviewer implies that a partial truth equals the whole truth. The same argument can also be applied to the additive partitioning if relative yield total or response ratio provides a kind of comparison between mixture and monocultures. Our statement is correct in the way that previous approaches are not designed to separate changes in ecosystem productivity by species interactions, as indicated by other reviewers. The additive partitioning is built on Price equation (covariance equation) that has never been biologically demonstrated for relevance in biodiversity partitioning (Bourrat et al., 2023).

We made clear that our work is built on and beyond the null expectation with addition of competitive expectation.

The manuscript relies on simulations because it claims that current experiments are unable to test this, given that they have replacement series designs (lines 128-131). There are, however, dozens of experiments where the replacement series was repeated at multiple densities, which would allow a direct test of these ideas. In fact, these ideas have already been tested in these experiments and density effects were found to be nonsignificant (e.g., Finn et al. 2013).

Out of point. Again, we are not testing density effect. Partial density is used to determine competitive growth responses that species may achieve in mixture based on their relative competitive ability. We used simulations, as partial density monocultures are used only in one experimental study that has been included in our study.

It seems that the authors are primarily interested in trees planted at a fixed density, with no opportunity for changes in density, and thus only changes in the size of individuals

(e.g., Fig. 1). In natural and experimental systems, realized density differs from the initial planted density, and survivorship of seedlings can depend on both intraspecific and interspecific interactions. Thus, the constrained conditions under which these ideas are explored in this manuscript seem narrow and far from the more complex reality where density is not fixed.

We use fixed density only for convenience. In biodiversity experiments, density can increase or decrease over time from initial levels. However, initial density is generally used in evaluation of species interactions. If interest is community productivity, density change does not need to be considered. Again, we are not testing density effects.

Additional detailed comments:

It is unclear to me which 'effects' are referred to on line 36. For example, are these diversity effects or just effects of competition? What is the response variable?

It means the effect of competitive interactions on productivity and should be clear based on previous sentences.

The usefulness of the approach is overstated on line 52. All partitioning approaches, including the new one proposed here, give the net result of many types of species interactions and thus cannot 'disentangle underlying mechanisms of species interactions.'

Not sure how many types of species interactions the reviewer referred to. If mechanisms of species interactions are grouped in three categories (positive, negative, and competitive) as has been in biodiversity research, our approach can tease them apart.

The weaknesses of previous approaches are overstated throughout the manuscript, including in lines 60-61. All approaches provide some, but not all insights. Sweeping statements that previous approaches are not effective, without clarifying what they can and can't do, is unhelpful and incorrect. Also, these statements imply that the approach proposed here addresses the limitations of these previous approaches. I don't yet see how it does so.

The weaknesses of previous approaches are not overstated in terms of separating changes in ecosystem productivity by species interactions. As pointed by other reviewers, none of the previous approaches are designed for quantifying changes in ecosystem productivity by species interactions.

The definitions given for the CE and SE on line 71 are incorrect. Competition affects both terms and CE can be negative or have nothing to do with positive interactions, as noted in many of the papers cited.

We are not trying to define CE and SE but only point out how CE and SE have been generally used in biodiversity research (see recent publication by Feng et al., 2022).

The proposed approach does not address the limitations noted on lines 73 and 74.

It does in terms of sources of net biodiversity effect, whether from positive, negative or competitive interactions.

The definition of positive interactions in lines 77 and 78 seems inconsistent with much of the literature, which instead focuses on facilitation or mutualism, rather than

competition when describing positive interactions.

Much of the literature supports our definition (see Loreau and Hector, 2001). In biodiversity research, positive interactions include resource partitioning and facilitation. What we are trying to point out is that competition affects species and community level assessments based on the null expectation and needs to be separated.

Throughout the manuscript, competition is often used interchangeably with resource competition (e.g., line 82) and complementarity is often attributed to resource partitioning (e.g., line 77). This ignores apparent competition and partitioning enemy-free niche space, which has been found to contribute to biodiversity effects in many studies.

If apparent competition refers to interference competition, it is included in negative interaction. Changes in species-specific pests and pathogens in mixture will be captured in positive or negative effects through facilitation or interference.

In what sense are competitive interactions positive for competitive species (lines 82-83)? By definition, competition is an interaction that has a negative effect. Do you mean that interspecific competition is less than intraspecific competition? I am having a very difficult time following the logic.

I am glad the reviewer raised this question that may confuse many others and has never been clearly discussed. It all depends on how comparison is made. If species performance in mixture are compared with that in partial density monocultures, as is in competition research, competition effect is negative for all species. If comparison is made between mixture and full density monocultures, as is done in biodiversity research, competition effect should be positive for more competitive species and negative for less competitive species, with resources flowing from less to more competitive species in mixture relative to full density monocultures.

Therefore, the definitions of competitive interactions based on additive series design in competition research cannot be used to describe competitive interactions based on replacement series design in biodiversity research. In biodiversity research, the effects of competitive interactions are never clearly defined at species or community level and mixed up with those of other species interactions.

Results are asserted on lines 93-95, but I cannot find the methods that produced these results. I am unable to evaluate the work without a repeatable description of the methods.

We have added references on sources of these data.

The description of the null hypothesis in the common additive partitioning approach on lines 145-146 is incorrect. In the null case, it does not assume that there are no interspecific interactions, but rather that interspecific and intraspecific interactions are equivalent.

Correct, changes have been made as suggested.

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

I recommend to:

- re-organize the presentation of the material (see my concerns in the public review section). The manuscript is very difficult to read.

Changes have been made to help with understanding of our approach. Figure 1 was modified to show the variations of competitive growth response with relative competitive ability from minimum (null expectation) to maximum (competitive exclusion).

- explore the mathematical form the the remainder term. It seems important to understand that the remainder capture terms unrelated to competition as defined in the present scope.

The remainder measures deviations from the null expectation, due to species differences in growth and competitive ability or competition effect. The term has clear meaning, positive for more competitive species and negative for less competitive species (lines 202-204), and does not need to be further explored or partitioned. The deviations of observed yields from competitive expectations are outlined in lines 205-221.

Reviewer #4 (Recommendations For The Authors):

The authors should be sure to include reproducible methods and share any data and code.

Both simulation and experimental data are shared through supplementary tables. Calculations are included in excel spreadsheets and do not require program coding.

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